Augmented Reality Experiment: Drivers' Behavior at an Unsignalized Intersection

Khaled F. Hussain, Essam Radwan, and Ghada S. Moussa

***Abstract*—Applying new technologies to traffic engineering studies has become more urgent due to the high cost and risk associated with ordinary in-the-field testing. Augmented reality (AR) is one of those technologies, in which virtual (computer-generated) objects are added to the real scene in a way that the user cannot distinguish between real and virtual objects in the final scene. Adding virtual objects (people, vehicles, hazards, and other objects) to the normal view can provide a safe realistic environment for testing driving performance under different scenarios. This paper presents two systems, i.e., AR vehicle (ARV) and offline AR simulator (OARSim) systems, and uses them to study the left-turn driving behavior at an unsignalized intersection for drivers with different characteristics. Two experiments were performed: one using the ARV system installed in a vehicle and another using the OARSim system installed in the laboratory. Quantitative measurements of left-turn drivers' behaviors were recorded. There was no significant gender effect on all measured parameters in both experiments. Older drivers selected larger gaps and used smaller acceleration rates to turn left than younger drivers in both experiments. The conservative driving attitude of older drivers indicates the potential presence of reduced driving ability of the elderly. While left-turn times using the ARV system were not significantly affected by drivers' age, older drivers took longer time to complete the left-turn maneuver than younger drivers using the OARSim did. Results from this study supported the feasibility and validity of the proposed systems and showed promise for these systems to be used as surrogates to in-the-field testing for safety and operation aspects of transportation research.**

***Index Terms*—Augmented reality (AR), drivers' behavior, intelligent transportation system, unsignalized intersection.**

I. Introduction

A USEFUL and very significant jump in simulation technology is to be able to evaluate synthetic simulated conditions in realistic settings [1]. This technology is based on augmented reality (AR). AR is a paradigm, which creates a combination of real and virtual objects in which the user cannot tell the difference between the real and augmented world [2]. Performing a real experiment to evaluate human performance under certain traffic scenarios might be very expensive, with a high degree of risk to drivers. The AR technology can offer a very realistic environment for enhancing drivers' performance on the road and testing drivers' ability to react to different road design and traffic operation scenarios. This can be achieved by adding virtual objects (people, vehicles, hazards, and other objects) to the normal view while driving an actual vehicle in a real environment. This makes applying the AR technology into traffic engineering applications a promising approach. For all AR applications, virtual objects must be aligned with real-world objects in the final view (see [1] and [3]).

In the U.S., about 70% of all annual fatalities due to vehicle crashes occur at unsignalized intersections [4]. Unsignalized intersections usually regulate low volume of traffic flow [5]. One significant safety concern at unsignalized intersections is the left turn across the path of opposite-direction traffic, where a left-turning vehicle is confronted by oncoming traffic [6]. In the U.S., the left turn across the path of opposite-direction traffic represents 27% of crossing path crashes at intersections [7]. These turning movements take place daily at unsignalized intersections, creating not only potential conflicts with the opposing traffic stream that is crossed but also potential delays for through traffic that shares the same traffic stream, particularly for two-lane roads [8].

Numerous studies have shown that drivers' characteristics (age and gender) are considered among the most important factors that affect left-turn drivers' behaviors at unsignalized intersections (other factors include driver distractions, opposing vehicles (type and speed), waiting time, time of day, and the intersection characteristics) [9]–[11]. Intersection crash involvement increases as the driver's age increases, and predominant crash types for older drivers are left-turning across oncoming traffic and crossing or turning into the traffic stream (see [12] and [13]).

Older drivers have problems with perception, cognition, and attention that can affect their ability to adequately detect, perceive, and accurately judge the safety of a gap when turning across traffic (see [10] and [14]). Therefore, older drivers may experience greater difficulties at unsignalized intersections. The National Highway Traffic Safety Administration reported that driving performance becoming progressively poorer with age accelerates somewhere in the fifth decade of life [15]. DeLucia *et al.* pointed out that judgments about whether a potential collision would occur were less accurate for older drivers (40–64 years old) compared with younger drivers (18–29 years old) [16]. In a driving simulator study at a stop-controlled intersection, involving three age groups (20–30, 31–55, and 56–83 years old), significant age and gender influences were

Manuscript received October 17, 2011; revised May 29, 2012, September 1, 2012, and October 9, 2012; accepted October 19, 2012. Date of publication November 15, 2012; date of current version May 29, 2013. This work was supported by the Center for Advanced Transportation Systems Simulation, University of Central Florida. The Associate Editor for this paper was A. Hegyi.

K. F. Hussain is with the Faculty of Computers and Information, Assiut University, Assiut 71516, Egypt (e-mail: khussain@aun.edu.eg).

E. Radwan is with the Department of Civil and Environmental Engineering, University of Central Florida, Orlando, FL 32816-2450 USA (e-mail: aeradwan@mail.ucf.edu).

G. S. Moussa is with the Department of Civil Engineering, Assiut University, Assiut 71516, Egypt (e-mail: ghada.moussa@gmail.com).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TITS.2012.2226239

1524-9050/$31.00 © 2012 IEEE

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

found. They indicated that the oldest group accepted larger gaps than the youngest two groups, and female drivers accepted larger gaps than male drivers [17]. They pointed out that older drivers and female drivers displayed a conservative driving attitude when making the left turn. An examination of a relative crash tendency of driver groups by age in left-turn crashes, using Kentucky crash data, showed that older drivers are more likely to be involved in left-turn crashes compared with younger drivers [18]. They indicated that, although females overall are more likely to be involved in a left-turn crash, older females are even more at a higher risk. Although older drivers accept larger gaps, these gaps may not be large enough to completely get rid of accident risk at unsignalized intersections [19]. However, other studies also indicate that the crash rates at unsignalized intersections are high for younger drivers [20].

The literature review highlighted that the left-turn across the path of opposite-direction traffic is a complex and risky driving maneuver, and is highly affected by left-turn drivers' characteristics (age and gender). Therefore, understanding the left-turn drivers' behavior under any given operation scenario can be an effective means of improving safety and operation at unsignalized intersections. This paper explores two different AR systems and applies them to study drivers' behaviors when conducting a left-turn maneuver across the path of opposite-direction traffic at an unsignalized intersection under different drivers' characteristics (age and gender). This paper, in addition to testing the feasibility of the application, tries to provide quantitative measurements of left-turn drivers' behaviors at unsignalized intersections.

II. Augmented Reality Systems' Discretions

Two AR systems are presented here: AR vehicle (ARV) system and offline AR simulator (OARSim) system. With the ARV system installed in a real vehicle, subjects are driving a real vehicle on a real safe road while seeing through a head-mounted display (HMD) an augmented video. The augmented video is a combination of a real-time video of the real surrounding road and virtual objects. On the other hand, with the OARSim system, subjects are taking control over the gas and brake pedals and the steering wheel (driving a simulator) while seeing through an HMD an augmented video. The augmented video is a combination of a prerecorded video of the real-world and virtual objects. Based on the drivers' application on brake and gas pedals and the steering wheel, the frame in front of his/her eyes changes accordingly.

The combination in both systems (ARV and OARSim systems), between the real-video and virtual objects, is done through the computer so that the driver is not able to tell the difference between real and virtual objects in the final scene. Detailed descriptions of both systems are presented in the following section, in addition to a comparison between in-the-field testing, testing using a driving simulator, and testing using these two systems.

A. ARV System

Our ARV system has three main components connected to a powerful computer. These components are a video camera, a flip-able HMD, and a GPS receiver unit. The ARV system can be installed in any vehicle, where the video camera is fixed on the vehicle's front windshield and the driver wears the HMD while driving the vehicle. Through the HMD, the driver will see an augmented video. The augmented video is a combination of a real-time video of the real surrounding road (from the video camera) and virtual (computer-generated) objects. The fixed video camera is aligned with the drivers' eye level to get the same drivers' prospective view of the real world. The fixed video camera gives the driver a view that enables him/her to drive easily and safely. Combining the real video with the virtual objects is done through the computer so that the driver will not be able to tell the difference between real and virtual objects in the scene.

The specifications of the ARV system are as follows.

1) The i-glasses Pro HMD has a Resolution of $800 \times 600 \times 3$, a field of view of 26° diagonal, a virtual image size of $70''$ at $13'$, a contrast ratio of $75:1$, a flicker-free refresh/display rate of 100 Hz, a full-stereo audio, and a weight of < 8 oz. It can be also adjusted to fit all individuals.
2) The GPS receiver unit used is a Navcom Starfire SF-2050 Ms GPS receiver combined with the Starfire satellite differential service and provides accuracy of 15 cm.
3) The computer used is a high-performance XPS Dell Laptop.
4) The camera has a video capture resolution of 640×480 pixels.
5) The 1500-W vector power inverter turns 12-V dc to 120-V ac.

The new generations of HMDs are much better than the first generation that had severe axis alignment problems, and user fatigue was common [21]–[23]. The type of HMD, which has been used in our system "Virtual i-O's i-glasses," delivers big-screen experience with a minimized risk of vertigo or motion sickness and can also be worn with eyeglasses. However, using HMDs for long duration might cause problems, even with new generations [24].

The block diagram that outlines the flow of data in the system is presented in Fig. 1. In which, the video camera takes a real-time stream video of the real scene. The GPS is attached to the vehicle to provide the location and the direction of the vehicle while it moves. The camera's intrinsic parameters (focal length, principle point, skew coefficient, and distortion coefficient) and extrinsic parameters (rotations and translations) need to be calculated. To calculate the camera's intrinsic parameters, the camera is calibrated offline. To calculate the camera's extrinsic parameters, two steps are needed. First, to calculate the camera's orientation, the camera is calibrated online using the unknown-orientation checkboard plan method once the system is installed in the vehicle and before its first use [25]. Second, the camera's translation is calculated using the GPS information (car position information) and an image processing algorithm. Both intrinsic and extrinsic parameters are used to align the virtual camera with the real camera. The graphic rendering uses the virtual camera position to render two 2-D frames: one that contains the virtual objects and the other that

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

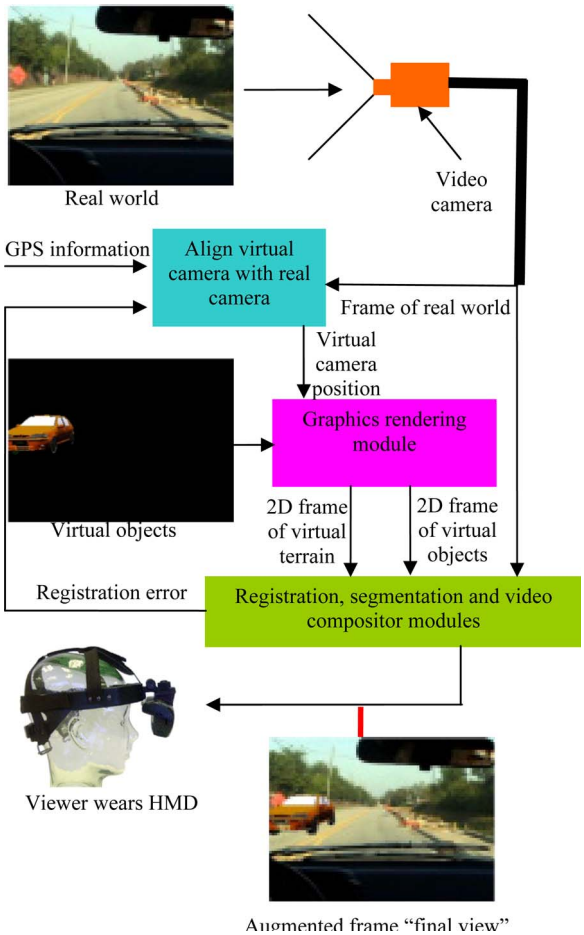

Fig. 1. Block diagram of the ARV system.

contains the virtual terrain. Then, the image registration module calculates the registration error between the real frame and the 2-D virtual terrain frame based on the affine transformation. This registration error is used to readjust the virtual camera. Then, the segmentation module segments the pixels that belong to the testing vehicle. Then, the video composition module combines the real frame and the virtual objects' frame to generate the augmented frame "final view." Finally, the final view is displayed in front of the viewer's eyes through the HMD. A detailed description of our system's intermediate steps is presented in [26].

B. OARSim System

The OARSim system consists of a high-definition (HD) video camera, a GPS receiver unit, an HMD, a driving wheel, brake and gas pedals, and a computer. The OARSim system can be divided into two main stages: recording and testing stages.

In the recording stage, the HD video camera is fixed on the vehicle's front windshield. The GPS is installed in a vehicle, and it is connected to a computer. The HD video camera records a video of the real world on its internal memory, and the GPS information is recorded on the computer during the journey of the vehicle.

In the testing stage, a driver sitting in the laboratory wears the HMD while taking control over the gas and brake pedals and the steering wheel. Through the HMD, the driver is able to

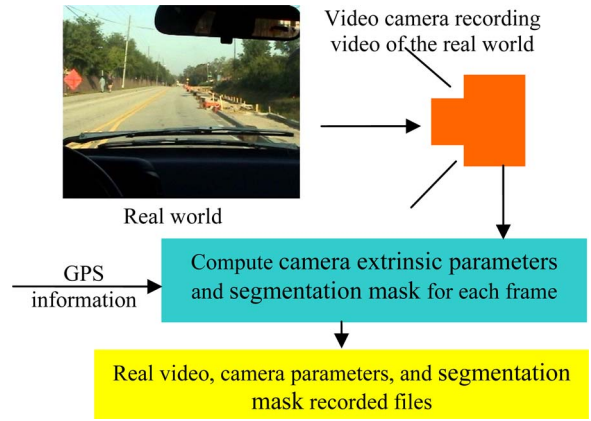

Fig. 2. Block diagram of the testing stage of the OARSim system.

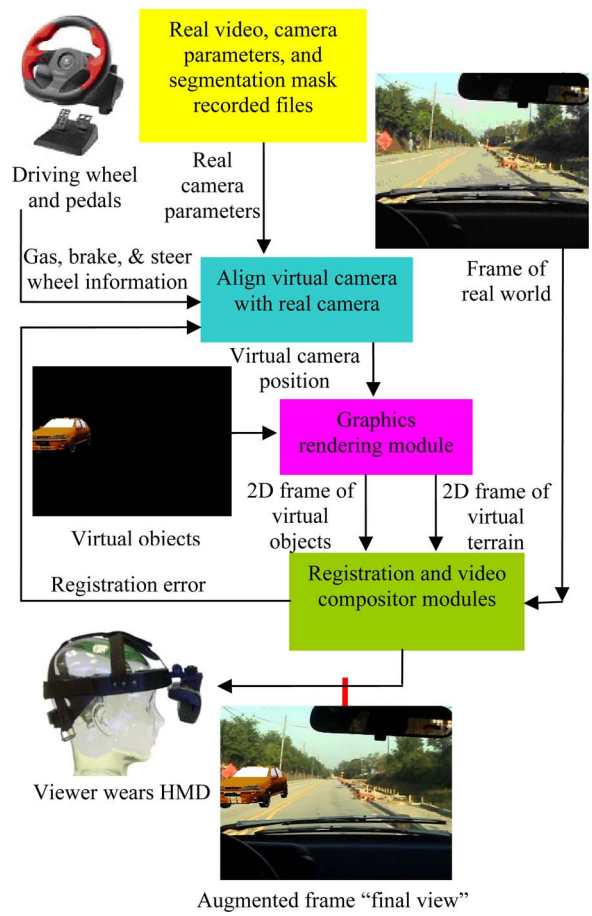

Fig. 3. Block diagram of the recording stage of the OARSim system.

see an augmented video. The augmented video is a combination of the prerecorded video, from the recording stage and virtual objects. This combination is done so that the driver is not able to tell the difference between real and virtual objects in the scene. Based on the drivers' application on brake and gas pedals and the steering wheel, the frame in front of his eyes changes accordingly.

In addition to the specifications of the ARV, the OARSim has a Logitech USB racing wheel with gas and brake pedals.

The block diagram that outlines the flow of the data in the OARSim system has two main parts: one for the recording stage and another for the testing stage, as outlined in Figs. 2 and 3,

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

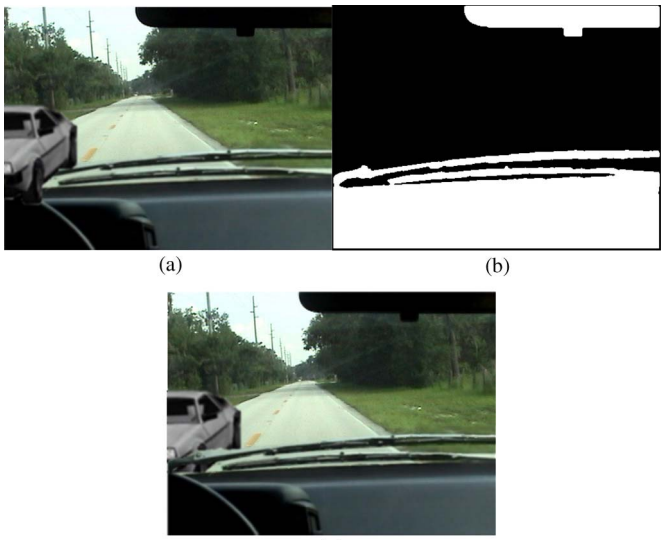

Fig. 4. (a) Visual error due to the insertion of a virtual vehicle into a prerecorded real frame without segmenting the testing vehicle. (b) Binary mask of the segmented testing vehicle. (c) Insertion of a virtual vehicle into a prerecorded real frame using the mask of the segmented testing vehicle.

respectively. In the recording stage, the HD video camera records a video of the real scene. The camera's parameters, both intrinsic and extrinsic, are calculated the same way as in the ARV system. In which, the intrinsic parameters (focal length, principal point, skew coefficient, and distortion coefficient) are calculated from camera calibration and extrinsic parameters (rotations and translations) are first calculated using the in-the-field marker and GPS data.

Applying the segmentation module is very important; otherwise, we may get a visual inconsistency frame, as shown in Fig. 4. The testing vehicle is segmented from the real frame based on the fact that the magnitude of the optical flow of the pixels that belong to the testing vehicle is approximately zero since these pixels do not move in general with respect to the camera. In calculating the optical flow, we used Horn and Schunck method [27]. The result of segmenting the testing vehicle from the real frame is a segmentation mask, as shown in Fig. 4.

Finally, the output of the recording stage is three files: 1) the recorded video file; 2) the camera transformation file that contains extrinsic parameters for each frame and intrinsic parameters; and 3) the segmentation mask file that contains a segmentation mask.

The output from the recording stage is used as input for the testing stage, as shown in Fig. 3. The information on camera parameters, gas and brake pedals, and the steering wheel are used to align the virtual camera with the real camera. Using the intrinsic parameters and adjusted extrinsic parameters of the virtual camera, the graphic rendering module renders virtual objects.

Consequently, the augmented frame "final view" is generated using the video composition module. If color values of the virtual objects' frame are combined with the color values of the prerecorded real frame (obtained from the recording stage) using a copy-and-paste approach, the result may not look natural because the real scene and virtual objects have different light conditions. Thus, we used Poisson image editing to create a smooth final image by blending the virtual objects' gradient into the real frame [28]. The modified virtual objects' colors are calculated as a solution of the following minimization problem for each color $c \in \{R, G, B\}$ separately:

$$\arg\min_{I^c} \iint_G F(\nabla I^c, \nabla g^c) \text{ with } I^c|_{\partial\Omega} = I_r^c|_{\partial\Omega} \quad (1)$$

where $F(\nabla I^c, \nabla g^c) = |\nabla I^c - \nabla g^c|^2 = ((\partial I^c/\partial x) - (\partial g^c/\partial x))^2 + ((\partial I^c/\partial y) - (\partial g^c/\partial y))^2$, I^c is the adapted color c of virtual objects, g^c is the original color c of virtual objects, I_r^c is the color c of the real frame, $\partial\Omega$ is the boundary of the virtual objects, Ω is the internal points of virtual objects, and $\nabla = [(\partial/\partial x), (\partial/\partial y)]$ is the gradient operator.

According to the variation principle, the solution of (1) must satisfy the Euler–Lagrange equation as follows:

$$\frac{\partial F}{\partial I} - \frac{d}{dx}\frac{\partial F}{\partial I_x} - \frac{d}{dy}\frac{\partial F}{\partial I_y} = 0. \quad (2)$$

The solution of the previous equation is the following Poisson equation with Dirichlet boundary conditions:

$$\Delta I^c = \Delta g^c \text{ over } \Omega, \qquad I^c|_{\partial\Omega} = I_r^c|_{\partial\Omega}. \quad (3)$$

To solve the Poisson equation (3), discretization approximation is used to compute the Laplacian as follows:

$$\begin{aligned}\Delta I^c &= \frac{\partial^2 I^c}{\partial x^2} + \frac{\partial^2 I^c}{\partial y^2} \\ &\cong \left(I_{x+1,y}^c - 2I_{x,y}^c + I_{x-1,y}^c\right) + \left(I_{x,y+1}^c - 2I_{x,y}^c + I_{x,y-1}^c\right) \\ &\cong I_{x+1,y}^c + I_{x,y+1}^c + I_{x-1,y}^c + I_{x,y-1}^c - 4I_{x,y}^c. \quad (4)\end{aligned}$$

Since I^c is unknown, (4) can be rewritten as follows:

$$\begin{pmatrix} 1 & 1 & -4 & 1 & 1 & 0 & \dots \\ 0 & 1 & 1 & -4 & 1 & 1 & \dots \\ 0 & 0 & 1 & 1 & -4 & 1 & \dots \\ 0 & 0 & 0 & 1 & 1 & -4 & \dots \\ 0 & 0 & 0 & 0 & 1 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ \vdots \\ x_N \end{pmatrix} = b \quad (5)$$

where $(x_1, x_2, \dots, x_N)$ are unknown N pixels (I^c), and b is a vector of length N representing Δg^c and the Dirichlet boundary conditions. Solution I^c is combined with the color values of the prerecorded real frame (obtained from the recording stage).

When the user steers the wheel, we cannot change the extrinsic parameters of the camera since the recorded video has been taken at specific extrinsic parameters. Therefore, to make the user to feel that the view point moves to the right or to the left, a subaugmented frame is displayed in front of the viewer's eyes through the HMD instead of the full frame. Let $s \in [-1, 1]$ be the input from the steering wheel. If s is equal to -1, it means that the steering wheel is turned to its maximum orientation toward left. If s is equal to 1, it means that the steering wheel is turned to its maximum orientation toward right. Whenever the user turns the wheel, s is changed. Then, the subaugmented frame is shifted horizontally accordingly

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

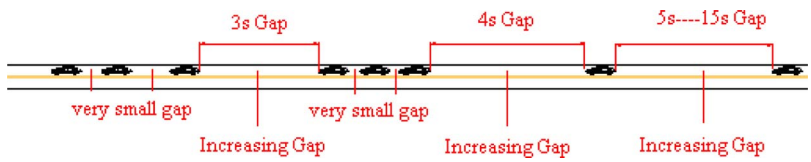

Fig. 5. Design of the virtual opposing traffic.

with amount $shh = ((w - w')/2)s$, where W is the width of the full frame, and W' is the width of the subaugmented frame.

The registration error between the prerecorded real frame (obtained from the recording stage) and the corresponding 2-D virtual terrain frame rendered using the approximated virtual camera parameters to render can be calculated using the registration module. This registration error is used to adjust the extrinsic parameters of the virtual camera.

III. Augmented Reality System Validation

In validating AR systems (ARV and OARSim systems), the systems' outcomes (final views) were considered, in which the systems' abilities to duplicate real scenes and to generate new scenes were assessed. By comparing real-world images with their corresponding augmented images "final view," we were able to assess the systems. As the augmented images are combination of a real view and virtual objects, it is very important to align real and virtual objects, in position, orientation, and scale, with each other. Small errors in this alignment generate visual inconsistencies, which can easily be detected by the user.

IV. Study of Drivers' Behavior at Unsignalized Intersection

Our ARV and OARSim systems were used to investigate drivers' behaviors when conducting a left-turn maneuver at an unsignalized intersection for different drivers' characteristics (age and gender). Two experiments were performed: one using the ARV system installed in a vehicle and another using the OARSim system installed in the simulation laboratory.

In the first experiment, using the ARV system, drivers drove a vehicle on a real safe two-lane road (a paved racetrack) while seeing, through the HMD, an augmented world. The augmented world is a combination of real-time video of the surroundings and computer-generated images.

In the second experiment, using the OARSim system, subjects were set in the laboratory, driving a fixed simulator (controlling a steering wheel and gas and brake pedals) while seeing through the HMD an augmented video. The augmented video is a combination of a prerecorded real video of a two-lane road, as in the first experiment, and computer-generated images, where the frame in front of the drivers' eyes changes according to his/her application on the brake and gas pedals and the steering wheel to ensure the faithfulness of the view.

In both experiments, one scenario was built, in which drivers drove on a straight segment until reaching a two-way stop-controlled intersection (unsignalized intersection) and were asked to select an appropriate gap between the oncoming virtual vehicles to make the left turn. During the scenario, the posted speed limit and the opposing virtual vehicles' speed were 25 mi/h (40 km/h).

TABLE I
Descriptive Statistics, by Age and Gender, of the Participants in the Final Study

Age	Gender	Mean	Standard deviation	N
Young (18-45)	Male	28.28	7.71	18
	Female	24.83	5.54	12
Old (45-65)	Male	55.13	4.55	8
	Female	52.33	4.18	6
Total	Male	36.54	14.36	26
	Female	34.00	14.25	18

To insure that drivers select their minimum acceptable gaps between opposing virtual vehicles and to make traffic appear random as in the real-world situation, the oncoming virtual traffic was formed in two classes of mixed gaps, as shown in Fig. 5. The first class was very small gaps that were unlikely to be accepted by left-turn drivers. The second class consisted of increasing gaps in which the succeeding gap was 1 s larger than the previous one. Ragland *et al.*, in their field study of gap acceptance and gap rejection behavior in the situation when one vehicle turns left across a path from the opposite direction, indicated that all gaps below 3 s were rejected and all gaps above 12 s were accepted [29]. Kettelson and Vandehey noted that it is obvious that most drivers will accept 15-s gaps [30]. Therefore, in this experiment, the very small gaps were less than 3 s, and the uniformly increasing gaps ranged from 3 to 15 s, to accommodate all drivers.

V. Experimental Procedure

Before left-turn experiments (using ARV and OARSim systems) began, participants were required to test-drive the systems to become familiar with them. Participants were informed that the objective of the study was to assess the fidelity of the AR systems and that they should obey traffic laws and rules when driving.

Eleven participants took part in the pilot study, with seven (five young and two older) making up the male group and four (three young and one older) covering the female group. The young group ranged from 18 to 45 years old, and the older group ranged from 45 to 65 years old. The results from the pilot study approved the design of all scenarios.

In the final study, 44 participants separated by gender and divided into two age groups participated in the study. Table I shows the descriptive statistics of participants, by age and gender, that participated in the final study.

All participants have a valid Florida's driving license for at least one year. After each scenario, each participant was given at least 2 min to rest before running the next scenario.

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

Fig. 6. Experiment using the ARV system.

Fig. 7. Experiment using the OARSim system.

Two experiments were conducted to study left-turn drivers' behavior at unsignalized intersection for different drivers' characteristics (age and gender) using the ARV system and the OARSim system.

Fig. 6 shows a picture of one of the participants during the experiment using the ARV system. In which, he is driving the vehicle on a paved racetrack while seeing through the HMD. A small picture of what he sees during the experiment is shown in the top left corner of Fig. 6. Another picture taken from outside the vehicle during the experiment is shown in the bottom left corner.

Another photo of another participant during the left-turn experiment using the OARSim is shown in Fig. 7. In which, she is driving the OARSim system while wearing the HMD. A small photo of what she sees during the experiment is shown in the top left corner of Fig. 7.

During the left-turn experiments, some parameters were recorded for each driver. Parameters included the actual size of the smallest time gap accepted by left turners (in seconds), the left-turn time (the time to complete the left-turning maneuver in seconds), and the left-turn acceleration (the average acceleration rate during the left-turn maneuver in feet per second squared).

To get a more subjective view regarding the realism of both systems (ARV and OARSim), a survey was administered to all drivers. The survey had two parts: The first part is related to the overall system fidelity, and the second part is related to the left-turn maneuver experiments. In the survey questions regarding the system fidelity, participants were asked about the scene visibility, scene realism, driving comfort, and HMD comfort. In additional, subjects were asked questions related to their feeling of driving, such as whether they felt any motion sickness while driving or not, and whether they felt safe while driving or not. In the survey questions regarding the left-turn maneuver, participants were asked about the left-turn maneuver comfort, scenario realism, and opposing vehicles' visibility and realism.

VI. Results and Interpretations

The results from validating the two systems (ARV and OARVSim) and the left-turn maneuver experiments are presented and discussed in this section as follows.

A. Systems' Validation

Although the ARV system used real-time videos and the OARVSim used prerecorded videos, the outcome from both systems (augmented view) was almost the same, which is a combination of real-world video and virtual vehicles. In validating AR systems' outcomes, the systems' abilities to duplicate real scenes and generate new scenes were assessed. The augmented view "final view" has to be free of any visual inconsistencies due to scale, orientation, and/or position errors in aligning visual objects with real objects in the final view.

In Fig. 8(a)–(d), two real-world images (a and b) and two augmented images (c and d) are shown. The augmentation in c and d appears visually very good without any visual inconsistencies, i.e., virtual objects (vehicles and fog) and real objects (road and trees) are well aligned. Comparing the augmented image in Fig. 8(c) with its corresponding real image in Fig. 8(b) indicates the system's ability to duplicate a real scene, clear of any visual problem. The duplicated scene is a combination of a real view and computer-generated objects (vehicles). In addition, comparing the augmented view in Fig. 8(d) with the real view in Fig. 8(a) shows the system's ability to generate new scene. This new scene is a combination of a real view and virtual objects (vehicles and fog).

B. Age and Gender Effects on the Left-Turn Maneuver

Figs. 9 and 10 show the average values with standard deviation bars of the recorded parameters (selected gap, left-turn time, and left-turn acceleration) for each age and gender group, during the ARV system experiment and the OARSim system experiment, respectively.

To statistically test if there is a significant gender or age effect on the recorded parameters in each experiment or not, an analysis of variance (ANOVA) was performed. ANOVA tests were performed at a confidence level of 95% ($\alpha = 0.05$).

1) Gap Acceptance: While the ANOVA result of both experiments (using ARV and OARSim) showed a significant effect of age on drivers' selected gap, the gender effect was not found to be significant. Old drivers in both experiments tend to accept larger gaps compared with younger drivers. The findings suggest that older drivers in both groups (males and females) are the most conservative drivers. This agrees with the findings in [17], [19], and [31].

2) Left-Turn Time: In the experiment using the ARV system, the ANOVA result showed that neither driver age nor

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

Fig. 8. Systems validation. Two real-world views: (a) with no vehicles and (b) with actual vehicles. Two augmented views: (c) with virtual vehicles and (d) with virtual vehicles and fog.

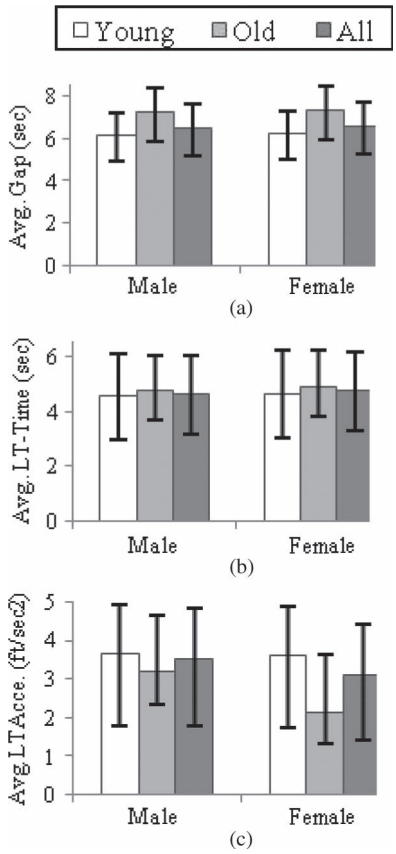

Fig. 9. Parameters of the left-turn MANEUVERS using the ARV system. (a) Averaged accepted gap. (b) Averaged left-turn time. (c) Averaged left-turn acceleration.

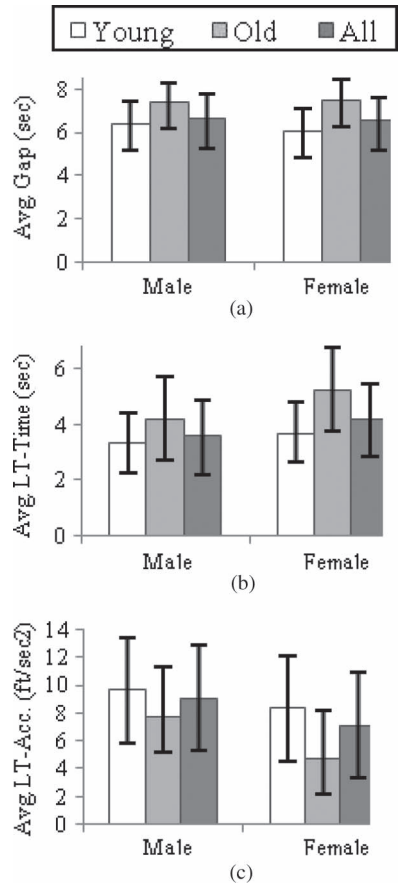

Fig. 10. Parameters of the left-turn MANEUVERS using the OARSim system. (a) Averaged accepted gap. (b) Averaged left-turn time. (c) Averaged left-turn acceleration.

gender has a significant effect on drivers' left-turn time. However, in the experiment using the OARSim system, older drivers took significantly longer time to complete the left-turn maneuver than younger drivers, but there was no gender effect. Staplin *et al.* pointed out that the difference in age between drivers did not significantly affect the left-turn time but the position of the vehicle prior to the left-turn movement did [32].

3) Left-Turn Acceleration: The ANOVA result showed that, in both experiments (using ARV and OARSim), only drivers' age has significant effect on drivers' acceleration rate during the period of the left turn at the intersection. During the process of turning left in the two experiments, older drivers are more likely to use smaller acceleration rates compared with the younger drivers. From both driving simulator research and field studies, it appears that the left-turning maneuver is done at smaller acceleration if the accepted gap is larger (see [17], [33], and [34]). Schaap *et al.* pointed out that it is not known whether the gap is selected for the desired acceleration or *vice versa* [34]. Moreover, the drivers' age effect on the acceleration rate is corresponding to the trend that older drivers accept larger gaps and can reflect older drivers' conservative driving attitude.

Based on the results, there was no significant gender effect on all left-turn parameters in both experiments. This is supported by the findings in [35] and [36].

C. ARV System versus OARSim System in Studying the Left-Turn Experiment

The average total time that it took drivers to run the left-turn experiment using the ARV system was 145 s, with a mininum

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

of 79.6 s and a maximum of 191.1 s. On the other hand, it took drivers to run the left-turn experiment using the OARSim system an average total time of 123.7 s, with a minimum of 57 s and a maximum of 171.1 s.

1) Gap Acceptance: In both experiments (using ARV and OARSim), all participants accepted gaps in a range of 4–9 s. The smallest gap that at least one participant accepted in both experiments was 4 s and the largest gap that at least one participant accepted was 9 s. The mean values of acceptable gaps were 6.55 and 6.59 s using ARV and OARSim, respectively. In both experiments, the 50th and the 85th percentile gap values were 7 and 8 s, respectively.

The Highway Capacity Manual (HCM) in 2000 defined the critical gap as the minimum gap that would be acceptable, i.e., all gaps that are equal or greater than the critical gap will be accepted and all the smaller gaps will be rejected [37]. The HCM recommends a base critical gap of 4.1 s, for a left turn from a major road, on both two-lane and four-lane major roads with a heavy-vehicle adjustment of 1 s for two-lane highways and 2 s for multilane highways [37]. However, the oversimplistic value of the HCM's critical gap may lead to inaccurate estimates of left-turn delay and capacity [38]. The American Association of State Highway and Transportation Officials (AASHTO) recommended a critical gap for left-turning from the major road across opposing traffic (Case F) for passenger cars, which is equal to 5.5 s, and to add an additional 0.5 s for each additional lane of travel to be crossed [39]. Moreover, Miscky and Mason measured critical gaps of 4.6 and 5.3 s for the 50th percentile gap values and 5.8 and 5.9 s for the 85th percentile gap values [40]. Field observations by Fitzpatrick *et al.* at stop-controlled intersections indicate that, for passenger cars, a 50th percentile gap of 6.5 s will be accepted (8.5 s for a five-axle truck) and an 85th percentile gap of 8.25 s will be accepted (10 s for a five-axle truck) [41]. Schalkwyk *et al.* pointed out that the 85th percentile values may be more appropriate than the 50th percentile values. He recommended that, in the case of larger older driver populations, a critical gap of 8 s is to be used [8].

2) Left-Turn Time: The mean values of the left-turn time using the ARV and the OARSim systems were 4.67 and 3.82 s, respectively. Fitzpatrick and Wolf noted that the AASHTO'1994 Green Book would represent a value of 4.3 s as the time to make the left-turn movement (for passenger car accelerating from a stop) [42]. Fitzpatrick and Wolf measured a value of 4.1 s for the travel time for left-turning vehicles at a rural T-intersection in Texas [42]. Moreover, Miscky and Mason measured the left-turn time at two intersections in Pennsylvania. They found the mean values of 4 and 4.3 s at the two intersections [40].

3) Left-Turn Acceleration: The mean values of the left-turn acceleration using the ARV and the OARSim systems were 3.34 and 8.26 ft/s^2, respectively. Glauz and Harwood indicated that the normal average acceleration is 4.69 ft/s^2 (1.43 m/s^2), which is the 65% of the maximum acceleration of the vehicle (7.22 ft/s^2) [43].

Student t-tests were conducted to test if there were statistical differences between the two experiments' recorded parameters (selected gap, left-turn time, and left-turn acceleration). Student t-tests were conducted at a 95% level of confidence $(\alpha = 0.05)$.

For accepted gap means, there was no significant difference between the first experiment (using the ARV system) and the second experiment (using the OARSim system). This might be because both systems (ARV and OARSim) provided drivers with almost the same augmented view, which is a combination of real-world video and virtual vehicles. Although the video used in the ARV system was a real-time video and that in the OARSim was a prerecorded video, this did not affect the drivers' accepted gap means.

On the other hand, there was a statistical significant difference in both left-turn time means and left-turn acceleration means between the two experiments. Drivers had a smaller left-turn time mean and a larger left-turn acceleration mean when using the OARSim system than when using the ARV system. Since both the left-turn time and the left-turn acceleration are attributed to the drivers' ability to push the gas pedal and to steer the vehicle wheel fast enough to turn, it appears that subjects had much better feeling and handling of both the gas pedal and the steering system of the ARV system than the OARSim system. This might be contributed to the nature of the portable simulator of the OARSim system that makes subjects believe that they are in a game and they tend to drive more aggressive.

D. Survey Analysis

In the survey questions regarding the system fidelity, participants were asked about the scene visibility, scene realism, scenario realism, driving comfort, and HMD comfort. Most of the answers indicated good overall system fidelity. None of the participants felt any kind of motion sickness as all scenarios were short-duration driving scenarios (less than 10 min each). All participants indicated that the experiment with the ARV system had none or small level of risk and no risk in the experiment with the OARSim system.

In the survey questions regarding the left-turn maneuver, participants were asked about the left-turn maneuver comfort and opposing vehicles' visibility and realism. Their answers in the two experiments showed an overall good level of comfort regarding the left-turn maneuver and an overall good visibility and realism of the opposing vehicles.

VII. Conclusion

AR is a paradigm that combines real-world video with virtual objects (computer-generated) in real time. Two systems based on the AR technology were presented: ARV and OARSim systems.

The ARV system can be installed in any vehicle in which the driver can wear the HMD while driving the vehicle. Through the HMD, the driver is able to see a combination of real video and virtual objects. While using the OARSim system, the driver wears the HMD while taking control over gas and brake pedals and a steering wheel. Through the HMD, the driver is able to see a combination of prerecorded real-world video and virtual objects. In both systems, the combination is done through the computer so that the driver cannot tell the difference between real and virtual objects in the scene.

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

The outcomes from the two systems (augmented views) were tested to be free of any inconsistency, virtual and real objects were aligned in position, orientation, and scale without any visual problems.

The two systems, i.e., ARV and OARSim, were used to study a left-turn maneuver at an unsignalized intersection. While there were no significant differences in the accepted gap means between the two experiments (using the ARV system and using the OARSim system), drivers had a smaller left-turn time mean and a larger left-turn acceleration mean when using the OARSim system than when using the ARV system. This might be because both systems (ARV and OARSim) provided drivers with almost the same augmented view (a combination of real-world video and virtual vehicles) that contributed to nonsignificant accepted gaps' mean difference. On the other hand, driving a nonreal vehicle (when using the OARSim system) had different effects from driving a real vehicle (when using the ARV system) on drivers' left-turn time mean and acceleration mean.

In the two experiments (using ARV and OARSim systems), there was no significant gender effect on all left-turn parameters (acceptable gap, left-turn time, and left-turn acceleration). On the other hand, there was a significant age effect on the accepted gap in both experiments. Both older males and older females needed bigger gaps than what was needed by the younger drivers.

Moreover, in the left-turn experiment using the OARSim system, older drivers took significantly longer to complete the left-turn maneuver than what the younger driver took, particularly in female groups. However, there was no age effect on the left-turning time in the left-turn experiment using the ARV system. Furthermore, in both experiments, older females had a smaller acceleration rate when performing the left-turn maneuver than younger females under the two systems.

Based on the survey questions that were handed to the participants, most of the participants indicated a good level of comfort, with none or a small level of risk. None of the participants felt any kind of motion sickness while driving. In addition, participants' answers indicated good visibility and realism of the scene, with overall good system fidelity.

This paper provided quantitative measurements of left-turn drivers' behaviors at unsignalized intersections. The findings of this paper can be useful for researchers and practitioners in the traffic engineering discipline. These findings may have an influence on the sight distance and capacity analysis of unsignalized intersections, which require further research investigation. Moreover, the results from this paper support the feasibility and validity of the proposed AR systems and they show promise for these systems to be used as surrogates to in-the-field testing for the safety and operation aspects of transportation research.

REFERENCES

[1] E. Gelenbe, K. F. Hussain, and V. Kaptan, "Simulating autonomous agents in augmented reality," *J. Syst. Softw.*, vol. 74, no. 3, pp. 255–268, Feb. 2005.

[2] K. F. Hussain and V. Kaptan, "Modeling and simulation with augmented reality," *Int. J. Oper. Res.*, vol. 38, no. 2, pp. 89–103, Apr. 2004.

[3] R. Woll, T. Damerau, K. Wrasse, and R. Stark, "Augmented reality in a serious game for teaching manual assembly processes," in *Proc. IEEE ISMAR-AMH*, Basel, Switzerland, 2011, pp. 37–39.

[4] L. Alexander, P. Cheng, M. Donath, A. Gorjestani, A. Menon, B. Newstrom, C. Shankwitz, N. J. Ward, and R. Starr, "Lag acceptance analysis for a rural unsignalized intersection," presented at the Proc. TRB 86th Annu. Meet. Compendium Papers, Washington, DC, 2007 (CD-ROM).

[5] J. Prasetijo and H. Ahmad, "Capacity analysis of unsignalized intersection under mixed traffic conditions," *Procedia—Social Behavioral Sci.*, vol. 43, pp. 135–147, 2012.

[6] C. Y. Chan, "Characterization of driving behaviors based on field observation of intersection left-turn across-path scenarios," *IEEE Trans. Intell. Transp. Syst.*, vol. 7, no. 3, pp. 322–331, Sep. 2006.

[7] D. L. Smith and W. G. Najm, "Analysis of crossing path crashes for intelligent vehicle applications," presented at the Proc. 8th World Congr. Intell. Transp. Syst., Sydney, Australia, 2001, Paper ITS00533.

[8] I. V. Schalkwyk and V. Stover, "Revisiting existing warrants for left-turn lanes at unsignalized intersections on two-way roadways," presented at the Proc. TRB 86th Annu. Meet. Compendium Papers, Washington, DC, 2007 (CD-ROM).

[9] D. G. Kim, Y. Lee, S. Washington, and K. Choi, "Modeling crash outcome probabilities at rural intersections: Application of hierarchical binomial logistic models," *Accid. Anal. Prev.*, vol. 39, no. 1, pp. 125–134, Jan. 2007.

[10] J. C. Laberge, J. I. Creaser, M. E. Rakauskas, and N. J. Ward, "Design of an Intersection Decision Support (IDS) interface to reduce crashes at rural stopcontrolled intersections," *Transp. Res. C, Emerg. Technol.*, vol. 14, no. 1, pp. 39–56, Feb. 2006.

[11] P. Hunt, D. N. Harper, and C. Lie, "Mind the gap: Training road users to use speed and distance when making gap-acceptance decisions," *Accid. Anal. Prev.*, vol. 43, no. 6, pp. 2015–2023, Nov. 2011.

[12] L. Hakamies-Blomqvist, "Fatal accidents of older drivers," *Accid. Anal. Prev.*, vol. 25, no. 1, pp. 19–27, Feb. 1993.

[13] S. Dissanayake and L. Perera, "A survey based study of factors related to older driver highway safety," *J. Transp. Safety Security*, vol. 3, no. 2, pp. 77–94, Jun. 2011.

[14] J. Oxley, B. Fildes, B. Corben, and J. Langford, "Intersection design for older drivers," *Transp. Res. F, Traffic Psychol. Behav.*, vol. 9, no. 5, pp. 335–346, Sep. 2006.

[15] "Addressing the safety issues related to younger and older drivers," U.S. Dept. of Transp., Washington, DC, DOT HS 807 957 NTS-31, 1993.

[16] P. R. DeLucia, M. Bleckley, L. E. Meyer, and J. M. Bush, "Judgments about collision in younger and older drivers," *Transp. Res. F, Traffic Psychol. Behav.*, vol. 6, no. 1, pp. 63–80, Mar. 2003.

[17] X. Yan, E. Radwan, and D. Guo, "Effects of major-road vehicle speed and driver age and gender on left-turn gap acceptance," *Accid. Anal. Prev.*, vol. 39, no. 4, pp. 843–852, Jul. 2007.

[18] S. Chandraratna and N. Stamatiadis, "Problem driving maneuvers of elderly drivers," *Transp. Res. Rec.*, no. 1843, pp. 89–95, 2003.

[19] J. Alexander, P. Barham, and I. Black, "Factors influencing the probability of an incident at a junction: Results from an interactive driving simulator," *Accid. Anal. Prev.*, vol. 34, no. 6, pp. 779–792, Nov. 2002.

[20] R. A. Retting, H. B. Weinstein, and M. Solomon, "Analysis of motor-vehicle crashes at stop signs in four US cities," *J. Safety Res.*, vol. 34, no. 5, pp. 485–489, 2003.

[21] M. Mon-Williams, J. P. Wann, and S. Rushton, "Binocular vision in a virtual world: Visual deficits following the wearing of a head-mounted display," *Ophthalmic Physiol. Opt.*, vol. 13, no. 4, pp. 387–391, Oct. 1993.

[22] S. Rushton, M. Mon-Williams, and J. P. Wann, "Binocular vision in a binocular world: New-generation head-mounted displays avoid causing visual deficit," *Displays*, vol. 15, no. 4, pp. 255–260, Oct. 1994.

[23] L. M. Dahlquist, K. E. Weiss, L. D. Clendaniel, E. F. Law, C. S. Ackerman, and K. D. Mckenna, "Effects of videogame distraction using a virtual reality type head-mounted display helmet on cold pressor pain in children," *J. Pedia. Psychol.*, vol. 34, no. 5, pp. 574–584, Jun. 2009.

[24] K. Ukai and P. A. Howarth, "Visual fatigue caused by viewing stereoscopic motion images: Background, theories, and observations," *Displays*, vol. 29, no. 2, pp. 106–116, Mar. 2008.

[25] J. Y. Bouguet, Open Camera Calibration Toolbox, Calif. Inst. Technol., Pasadena, CA. [Online]. Available: http://www.vision.caltech.edu/bouguetj/calib_doc/

[26] G. Moussa, E. Radwan, and K. Hussain (2012, Apr.). Augmented reality vehicle system: Left-turn maneuver study. *Transp. Res. C, Emerg. Technol.* [Online]. *21(1)*, pp. 1–16. Available: http://www.sciencedirect.com/science/article/pii/S0968090X1100115X

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.

[27] B. K. P. Horn and B. G. Schunck, "Determining optical flow," *Artif. Intell.*, vol. 17, no. 1-3, pp. 185–203, Aug. 1981.
[28] P. Pérez, M. Gangnet, and A. Blake, "Poisson image editing," *ACM Trans. Graph.*, vol. 22, no. 3, pp. 313–318, Jul. 2003.
[29] D. R. Ragland, S. Arroyo, S. E. Shladover, J. A. Misener, and C. Y. Chan, "Gap acceptance for vehicles turning left across oncoming traffic: Implications for Intersection Decision Support design," presented at the Proc. TRB 85th Annu. Meet. Compendium Papers, Washington, DC, 2006 (CD-ROM).
[30] W. K. Kettelson and M. A. Vandehey, "Delay effects on driver gap acceptance characteristics at two-way stop-controlled intersections," *Transp. Res. Rec.*, vol. 1320, pp. 154–159, 1991.
[31] N. D. Lerner, R. W. Huey, H. W. McGee, and A. Sullivan, "Older Driver Perception-Reaction Time for Intersection Sight Distance and Object Detection," Federal Highway Admin., Washington, DC, Rep. FHWA-RD-93-168, 1995.
[32] L. Staplin, D. Harkey, K. Lococo, and M. Tarawneh, "Intersection Geometric Design and Operational Guidelines for Older Drivers and Pedestrians," Federal Highway Admin., Washington, DC, Final Rep. Pub. FHWA-RD-96-132, 1997, vol. 1.
[33] A. Polus, "Gap acceptance characteristics at unsignalized urban intersections," *Traffic Eng. Control*, vol. 24, no. 5, pp. 255–258, May. 1983.
[34] T. W. Schaap, T. V. Dijck, B. V. Arem, and P. Morsink, "A PC microsimulation of a gap acceptance model for turning left at a T-junction.," in *Proc. TRB 88th Annu. Meet.*, Washington, DC, 2009, pp. 1–11.
[35] K. H. E. Kroemer and H. B. Kroemer, *Ergonomics: How to Design for Ease and Efficiency*. Englewood Cliffs, NJ: Prentice-Hall, 1994.
[36] R. J. Koppa, *Human factors-from Traffic Flow Theory*, N. Gartner, C. J. Messner, and A. J Rathi, Eds. Washington, DC: Transportation Research Board, 1992, ch. 3.
[37] *Highway Capacity Manual*, 4th ed., Transp. Res. Board, Washington, DC, 2000.
[38] P. Devarasetty, Y. Zhang, and K. Fitzpatrick, "Differentiating between left-turn gap and lag acceptance at unsignalized intersections as a function of the site characteristics," *J. Transp. Eng.*, vol. 138, no. 5, pp. 580–588, May 2012.
[39] *A Policy on Geometric Design of Highways and Streets*, 5th ed. Washington, DC: Amer. Assoc. State Highway Transp. Officials (AASHTO), 2004.
[40] R. J. Miscky and J. M. Mason, "Sight distance for vehicles turning left off major roadway," in *Proc. Nat. Conf. Access Manag.*, Vail, CO, 1996, pp. 353–382.
[41] K. Fitzpatrick, J. Mason, Jr., and D. Harwood, "Comparison of sight distance procedures for turning vehicles from a stop-controlled approach," presented at the Proc. Annu. TRB Meet., Washington, DC, 2005.
[42] K. Fitzpatrick and T. Wolf, "Left-turn lane installation guidelines," in *Proc. 2nd Urban St. Symp.Transp. Res. Board*, Anaheim, CA, 2003, pp. 1–22.
[43] W.D. Glauz and D. Harwood, *Vehicles. Traffic Engineering Handbook*, 5th ed. Washington, DC: Inst. Transp. Eng., 1999, ch. 3, pp. 50–77.

Khaled F. Hussain received the B.Sc. (with distinction and honors) and M.S. degrees in electrical engineering, both from Assiut University, Assiut, Egypt, in 1994 and 1996, respectively, and the Ph.D. degree in computer science from the University of Central Florida, Orlando, in 2001.

From 2002 to 2006, he worked as a Visiting Assistant Professor with the University of Central Florida. From 2006 to 2007, he worked as an Assistant Professor with the Department of Electrical Engineering, Faculty of Engineering, Assiut University. Since 2007, he has been with the Department of Computer Science, Faculty of Computers and Information, Assiut University, and is currently an Executive Director with the Multimedia Laboratory, Faculty of Computers and Information, Assiut University. His major research interests include computer vision, computer graphic, augmented reality, and computer animation.

Essam Radwan received the B.S. degree in civil engineering (with honors) from Cairo University, Giza, Egypt, and both the M.S. and Ph.D. degrees from Purdue University, West Lafayette, IN.

In 1978, he started his teaching career with Virginia Tech, where he was tenured and promoted to Associate Professor. In 1984, he moved to Arizona State University and was promoted to Full Professor in 1990. Through a national search, he was recruited to the University of Central Florida (UCF), Orlando, to chair the Department of Civil and Environmental Engineering, and he remained in that position until 2003. In 1998, he was appointed founding Director of the Center for Advanced Transportation Systems Simulation (CATSS), UCF. He is currently an Interim Chair with the Department of Civil, Environmental, and Construction Engineering, College of Engineering and Computer Science, UCF. He also serves as an Executive Director with CATSS. The "Human Centered Transportation Simulation Program" that he created under CATSS auspices has been viewed as unique and one of the leading programs in the U.S. He directed and codirected about 55 research projects, totaling well over $11 million in external funded research projects. Through research and graduate advising, he has published more than 250 technical papers and reports, and his work has been extensively cited by his peers in the U.S. and around the world. His research interests include traffic characteristics and traffic signal control of freeway and street intersections, particularly theoretical and applied models, discrete-event simulation models, and large-scale databases developed to describe traffic behavior.

Professor Radwan is a Fellow of the American Society of Civil Engineers (ASCE) and the Institute of Transportation Engineers and has been an active member of the Transportation Research Board, the Florida Engineering Society, and the Intelligent Transportation Systems of Florida. He has received numerous honors and awards, including the ASCE Frank Masters Award, the UCF University Excellence in Professional Service, the Office of Sponsored Research and Commercialization UCF Millionaire's Club Member designation (UCF faculty who attracted outside research funding exceeding $1 million) for three consecutive years, the Florida Engineering Society Outstanding Technical Achievement Award, the UCF Distinguished Research of the Year Award, and the UCF Faculty Excellence in Mentoring Doctoral Students Award.

Ghada S. Moussa received the B.S. degree in civil engineering (with honors) from the University of Assiut, Assiut, Egypt, in 2000 and both the M.S. degree in structural engineering and the Ph.D. degree in transportation engineering from the University of Central Florida, Orlando, FL.

Since 2006, she has been an Assistant Professor with the Department of Civil Engineering, Faculty of Engineering, Assiut University. She teaches courses related to transportation engineering, traffic engineering, and computer applications in traffic operation. Her major research interests include the application of computers into traffic engineering studies to enhance the operation and safety of highways and streets.

Authorized licensed use limited to: Politechnika Krakowska. Downloaded on March 22,2026 at 12:35:31 UTC from IEEE Xplore. Restrictions apply.